

Chapter 1 of 5 — A Letter Across a Country

It begins with an experiment. In the 1960s, letters were mailed to strangers in Nebraska. Each one carried a single rule: pass it only to someone you know by first name. Each one had a single target, a stockbroker in Boston. How many hands does a letter like that need? My guess is below, next to what actually happened.

 No description has been provided for this image

Milgram's letter experiment (1960s)

Read the picture as a relay. A packet starts with a randomly chosen person in Omaha, Nebraska, and has to reach one named stockbroker near Boston. Nobody may post it directly: each holder passes it to a single person they know on a first-name basis and who they think sits closer to the target. Each arrow is one such hand-off. The drawing shows the idea, not the result.

🌐 Six degrees of separation

In the 1960s, Stanley Milgram mailed packets to strangers in Nebraska with one rule: pass it to someone you know on a first-name basis, until it reaches a Boston stockbroker. The letters that arrived took about **6 hops** — 64 of the 296 packets made it, through **5.2 intermediaries** on average. It has been re-measured twice since: a 2003 email version (24,000 chains aimed at 18 targets in 13 countries) averaged about **4 steps** among the chains that finished, and a 2012 study of Facebook's 721 million users found **4.74 steps**.

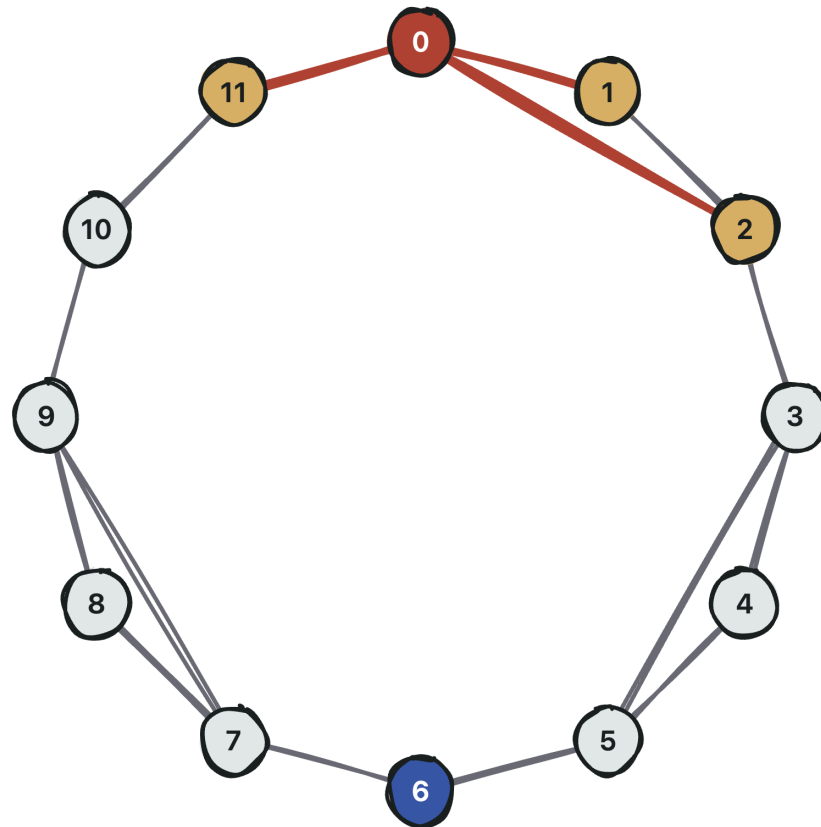
Try it: Wikirace

▶ wiki-race.com

Opens in a new tab, so this page stays where it is. The game hands you a **start** article and a **target** article. You have to reach the target by clicking. You may only use the links printed on the page you are standing on. No search box, no typing an address. Play a round if you feel like it, and keep count of your clicks.

The letter-holder's view

The **rust** dot is whoever holds the letter right now. The **rust lines** are the only friendships they can see: the ones running to their own **amber** friends. The **navy** dot on the far side is the Boston target. Every **grey** dot and line exists, but the holder has no idea who out there knows whom. They pick ONE amber friend and hope. Dots are drag-able.



Existing vs. findable

A short path **existing** somewhere in a network is a fact about the whole network — you'd need the full map to even check it. A short path being **found**, by ordinary people each using only who THEY personally knew, is a much stronger claim. Milgram's letter-holders never saw the social network as a whole; they just kept forwarding to whoever seemed "closer" to the target. That they succeeded, in about 6 hops, is the surprising part.

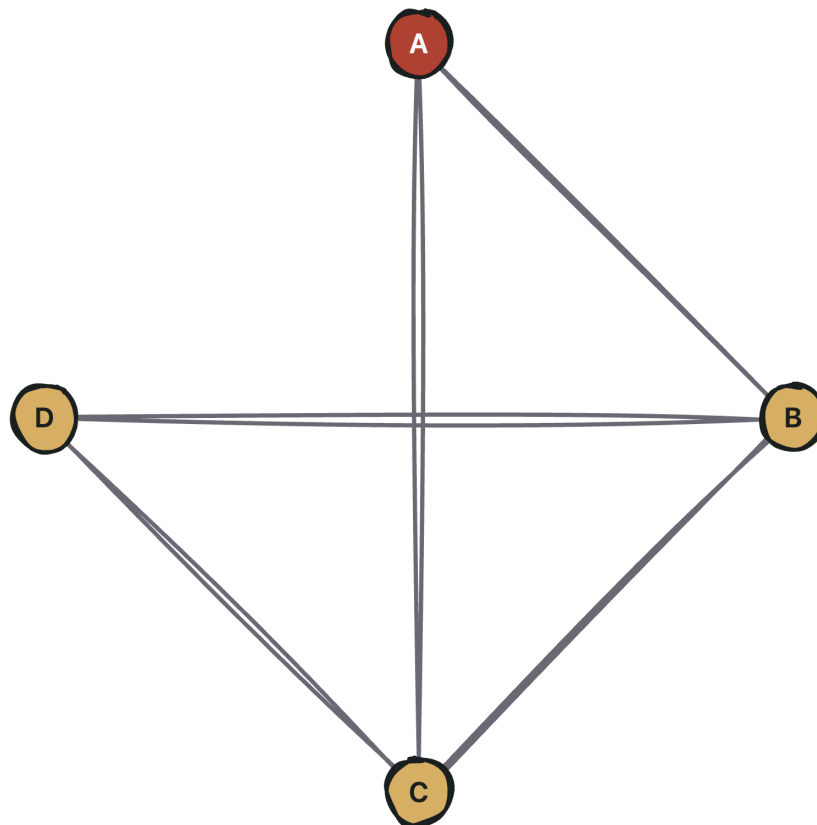
Chapter 2 of 5 — Measuring Smallness

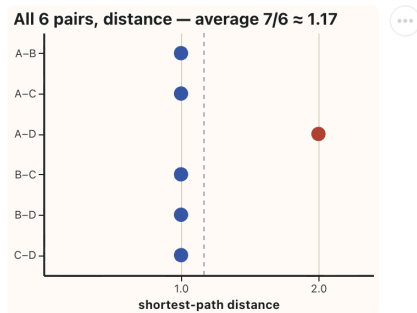
"Small" needs a number before it means anything. This chapter builds the module's measuring sticks. The first is distance: how many steps separate two people. We find it on a four-person network, and the same chart hides a second number, the worst-case trip. Then distance again, by hand, on paper. After that comes a family of measures distance cannot see at all. They are about the shape of one person's circle of friends. One person's own corner first. Then everyone's corners averaged. Then the whole network counted a third way, which does not agree with the second.

Steps from A 

Wave from A — 3 step(s)

Four people, five friendships. **A** is rust. **Amber** dots are everyone A can reach in that many steps or fewer. **Grey** dots are still out of range. Drag the slider up one notch at a time and watch who joins. The step at which someone first turns amber is their distance from A. You can drag the dots too; moving them changes nothing but the picture.





One dot per pair of people. With four people there are exactly six pairs, and every pair gets counted once. A dot's position along the axis is that pair's distance. A dot at 1 means that pair sits at distance 1. The red dot is the pair that sits at distance 2. The dashed line is at the average of the dots, which is the average for the state.

Distance and average path length

The **distance** $d(u, v)$ between two people is the number of edges on the shortest route between them. Averaging over every pair gives the **average path length** L — one number for how far apart a whole network sits:

$$L = \frac{\text{every pair's } d(u, v), \text{ added up}}{\text{how many pairs there are}} = \frac{1 + 1 + 2 + 1 + 1 + 1}{6} = \frac{7}{6} \approx 1.17$$

Five pairs sit at distance 1 here and A–D at 2. "Six degrees" is exactly this number, measured on a whole country.

Diameter — the worst-case trip

Average path length (L) is the typical trip; the **diameter** is the single LONGEST shortest-path trip anywhere in the network — the worst case, not the average. Here every pair sits at distance 1 except A–D, so the diameter is **2**. A ring or lattice can have a diameter that keeps growing as the network grows; a small-world network's diameter stays small even as it gets bigger — that's part of what "small" means.

To re-upload:
Photo of your paper work

Uploaded 1 file. [Click to clear file.](#)

Drop a photo of your paper here. Show the 5 dot ring, all 10 pairs with a distance beside each, and the average at the bottom. Marking shows keeps a tidy answer.

This is exactly what your tutor will see. Missed a pair, or want to redo the table? Drop another photo into the box above — it replaces this one, as many times as you like. When it looks right, press send.

 Send to

✓ **Sent.** Your tutor is looking at it now.

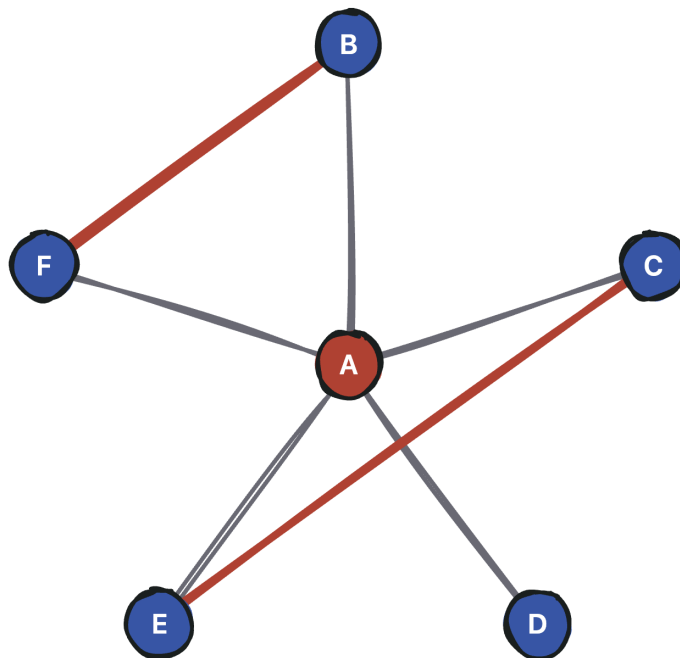
By hand: distance on a 5-ring

Same two ideas as before — $d(u, v)$, the distance between two people (the fewest lines you cross to get from u to v), and L , the average of $d(u, v)$ over every pair — computed entirely on paper this time, on a network no widget ever showed. 5 pairs sit at distance 1 and 5 "across the circle" pairs sit at distance 2:

$$L = \frac{5 \times 1 + 5 \times 2}{10} = \frac{15}{10} = 1.5$$

A and A's five friends

A is the rust dot in the middle; B, C, D, E and F are its five friends. The grey lines from A are A's own friendships — ignore those. The **rust lines run between two of A's friends**: those are the ones that say "my friends know each other". Find them and count them yourself. Dots are drag-able if the picture gets tangled.



👉 Local clustering — do my friends know each other?

The **local clustering coefficient** of a person i counts the friendships among i 's own friends, against how many there could be:

$$C_i = \frac{\text{friendships among } i\text{'s friends}}{\text{pairs of friends } i \text{ has}} \implies C_A = \frac{2}{10} = 0.2$$

A has 5 friends, so up to 10 friendships could exist among them; 2 do. That is one person's score; everybody in the picture has one of their own, and social networks typically run high.

👥 Average local clustering — one vote each

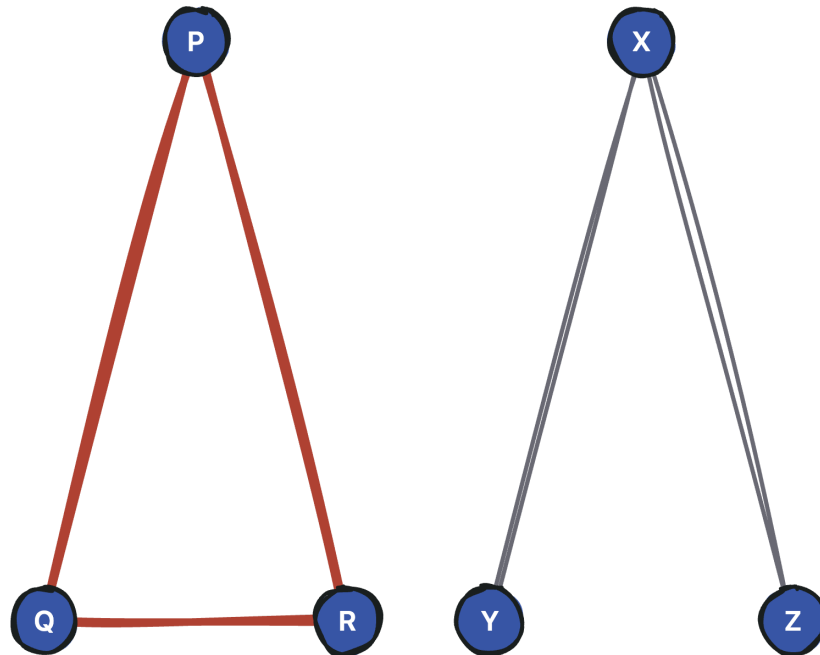
Everyone has their own C_i — the fraction of their friend-pairs who are friends. The **average local clustering** $\bar{C}$ is the plain mean of those over everybody. A scores 0.2; B, C, E and F each know exactly two people who know each other, so each scores a perfect 1; D, with a single friend, has no pair to check at all — by convention anyone with fewer than two friends scores 0. So

$$\bar{C} = \frac{\text{everyone's } C_i, \text{ added up}}{\text{how many people there are}} = \frac{0.2 + 1 + 1 + 0 + 1 + 1}{6} = \frac{4.2}{6} = 0.70$$

Everyone gets one equal vote, which is how four small, perfectly closed circles outvote the one hub whose own circle is nearly empty.

Closed triplet (left) — open triplet (right)

Two trios. Left: P, Q and R, with **all three friendships there (rust)**. That is a **closed** triplet, better known as a triangle. Right: X knows Y and Z, but Y and Z never met. Still a triplet, just **open**: one line short of a triangle. A triplet is named by its middle person, which on the right is X. In a triangle every corner takes a turn in the middle. That is why one triangle shows up three times in a triplet count. Dots are drag-able.



▲ Global clustering — one number for the whole network

A **triplet** is three people joined by at least two friendships, counted by the person in the middle — open and closed alike. A **triangle** is a triplet whose third friendship also exists. Someone with k friends (k = how many friends that person has) centers $k(k-1)/2$ triplets: Alice ($k = 5$) centers 10, her other four friends center 1 each, D none — 14 in all, holding 2 triangles. The **global clustering coefficient** is

$$C = \frac{3 \times \text{triangles}}{\text{triplets}} = \frac{3 \times 2}{14} = \frac{6}{14} \approx 0.43$$

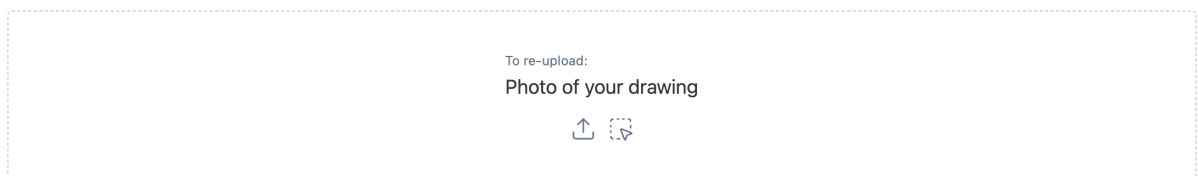
— the 3 because each triangle is counted once per corner. People who center more triplets get more say, so this number listens to hubs. **Same network, two answers:**

$$\overline{C} = 0.70 \quad \text{versus} \quad C = 0.43$$

Both are correct. The average counts PEOPLE, and four of the six here have tiny, perfectly closed circles that each score 1. The global one counts TRIPLETS, and Alice alone owns 10 of the 14 — only 2 of them closed. Reporting either as "the clustering coefficient" without saying which is how two papers end up impossible to compare.

Chapter 3 of 5 — One Extra Line

A ring of dots, and one extra cable to spend anywhere. Where should it go? That one choice opens this chapter's real question: what a single link can do to a whole network. From there we work out two formulas: how a ring's clustering behaves as the ring grows, and how its distances do.



Uploaded 1 file. [Click to clear file.](#)

Drop a share photo of your 8 dot ring here — the drawing with your one extra cable on it. It does not need to be neat; it needs to show which two dots you joined.

This is exactly what your tutor will see. Not the one you meant? Drop another photo into the box above — it replaces this one, as many times as you like. When it looks right, press send.

 **Send to**

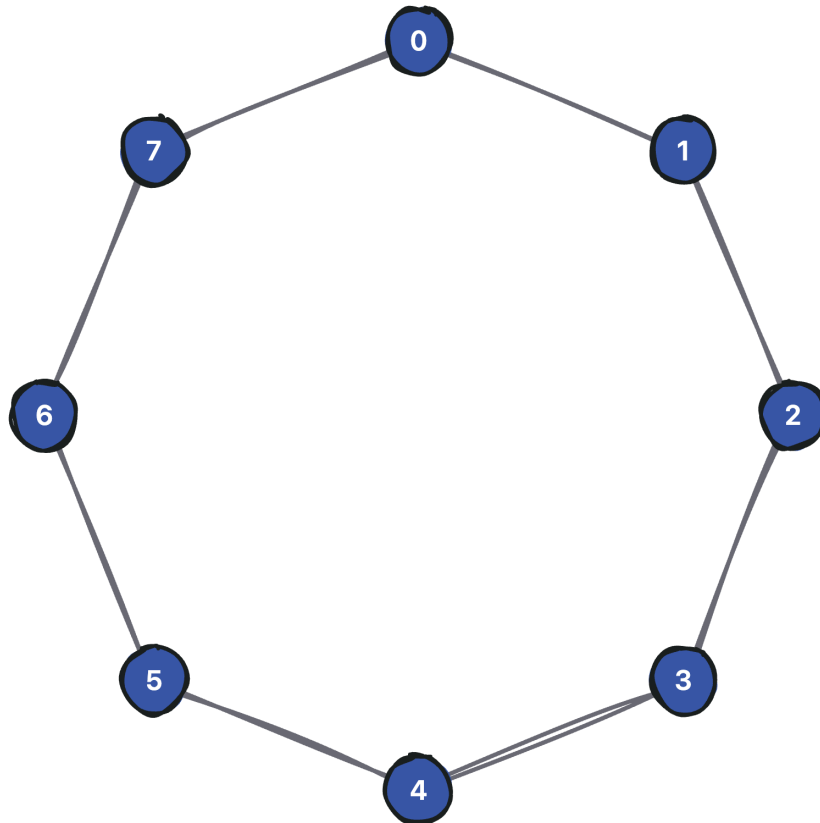
✓ **Sent.** Your tutor is looking at it now.

Compare three placements

- ☒ no extra cable
- ☐ short cable (2 apart)
- ☐

Average distance: 2.29

The same 8-dot ring three times over. The **rust line is the one extra cable**; the number above is the average trip length over all 28 pairs, recomputed every time you switch. Flip between the three options and watch only that number: same ring, same single extra line, and the only thing that changed is where it was put.



🔌 One cable, two very different effects

Same single line, two jobs. Between nearby dots it closes a **triangle** — clustering goes up. Across the ring it is a **shortcut** — trips get shorter for everyone at once. On this 8-dot ring the average distance goes $2.29 \rightarrow 2.07$ with a short cable and $2.29 \rightarrow 1.96$ with a long one: about half again as much travel saved, from one line placed differently. WHERE a link goes matters more than HOW MANY links there are — and the bigger the ring, the wider that gap gets.

k (friends per person)   check my count (highlight friendships among the amber dots)

Node 0 is the rust dot; the **amber dots are its friends** (2 on each side). Every line is a friendship — the ones to count are the lines that join *two amber dots*. The slider changes how many friends each person has. The check-box paints those amber-to-amber lines rust once you have committed to a number, so you can check your own count.

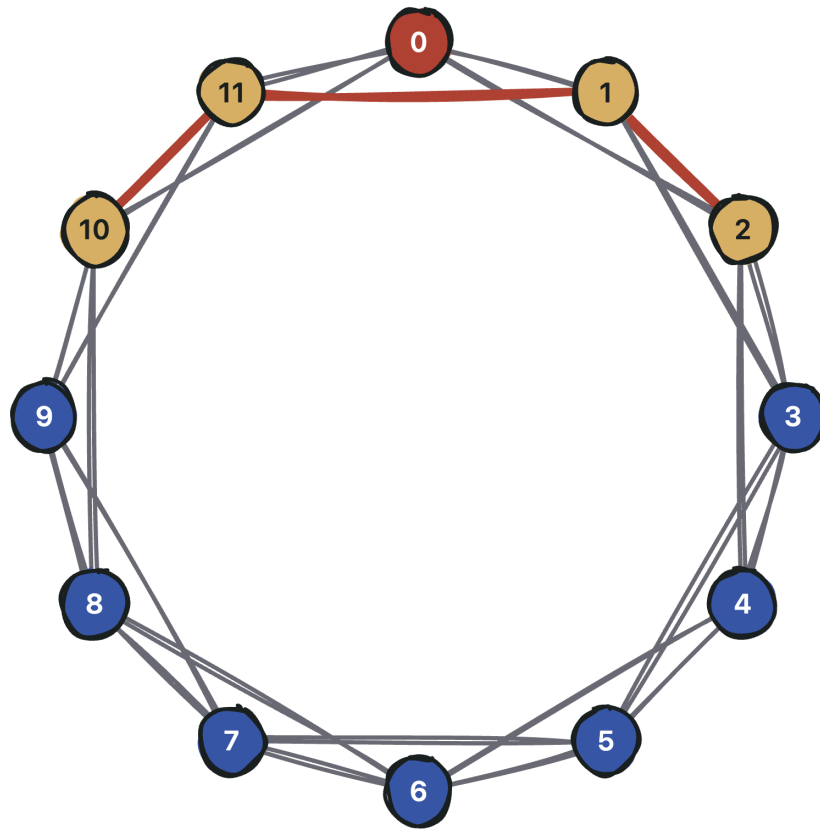


Photo of your ring working (triangles + formulas)



Drop a phone photo of your derivation here. Show node 0's friends, which pairs among them already know each other, and the two formulas you ended up with. Crossings-out are fine and welcome.

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My own work on paper — the task: On a 12-dot ring with friend-count k , work through node 0's friends and clustering, reason that clustering depends only on k not N , work out hops across a 1000-person ring, and relate average path length L to N and k .

▲ Clustering and path length on a ring — as formulas

N = people in the ring, k = friends each person has ($k/2$ per side), C = clustering (how often two of your friends know each other), L = average distance (typical steps between two people).

$$C(k) = \frac{3(k-2)}{4(k-1)} \qquad L(N, k) \approx \frac{N}{2k}$$

Clustering is a function of k ONLY, never N ($k = 2$ gives $C = 0$, $k = 4$ gives $C = 0.5$). Every neighbourhood is identical, so counting around one node gives the whole ring. L grows with the crowd instead: the farthest trip is about N/k hops — half a ring at $k/2$ places per hop — and the average is about half of that.

The puzzle: clustered AND close?

Same four letters as before: C = clustering (how often two of your friends know each other), L = average distance (typical number of steps between two people), N = how many people there are, k = how many friends each person has. **Ring world:** tight communities (high C , independent of N) but journeys that get longer and longer as N grows ($L \approx N/(2k)$). **Random world:** short journeys (low L) but no community (low C). This world has formulas too — throw everyone's k friendships at random and both come out in closed form:

$$C_{\text{random}} \approx \frac{k}{N} \qquad L_{\text{random}} \approx \frac{\log N}{\log k}$$

Those two are clustering and average distance measured on the random world. The first is one hundredth when a thousand people have ten friends each. The second is just "how many times do I multiply by k before I have covered all N people" — with ten friends each, 3 steps for a thousand people, 6 for a million, 9 for a billion. **Short paths are free.** Anything well mixed already has them, so the surprising half of six degrees was never the six — it is having six AND keeping the triangles. Real social networks somehow manage both, and neither extreme world explains it.

Chapter 4 of 5 — Turning One Dial

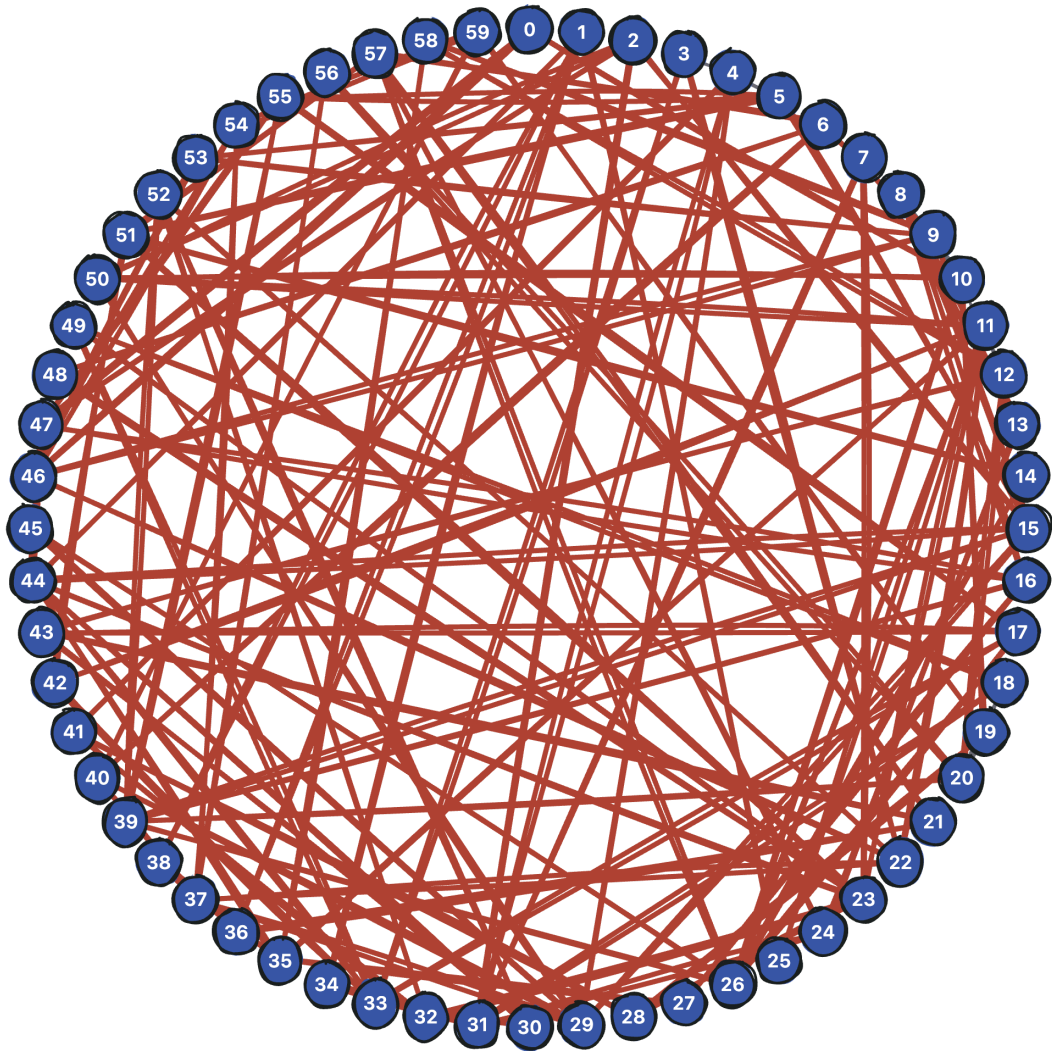
Chapter 3 ended in a tension. The ring world is clustered but far-flung; the random world is close but has no community. This chapter puts a dial between the two. Rewire a fraction p of the ring's links to random targets. Then watch what each measure does as the dial turns: first on a slider, then with real code at $N=2000$.

Reading the dials: p = the fraction of the ring's links picked up and reconnected to someone chosen at random. That is the slider below. L = average distance, how many steps apart a typical pair is. C = clustering, how often two of your friends know each other. L_0 and C_0 are those same two numbers for the untouched ring ($p = 0$). So $L/L_0 = 1$ means "unchanged", and $L/L_0 = 0.2$ means distances shrank to a fifth.

rewiring probability 

$p = 1.0$ distance $L/L_0 = 0.16$ clustering $C/C_0 = 0.02$

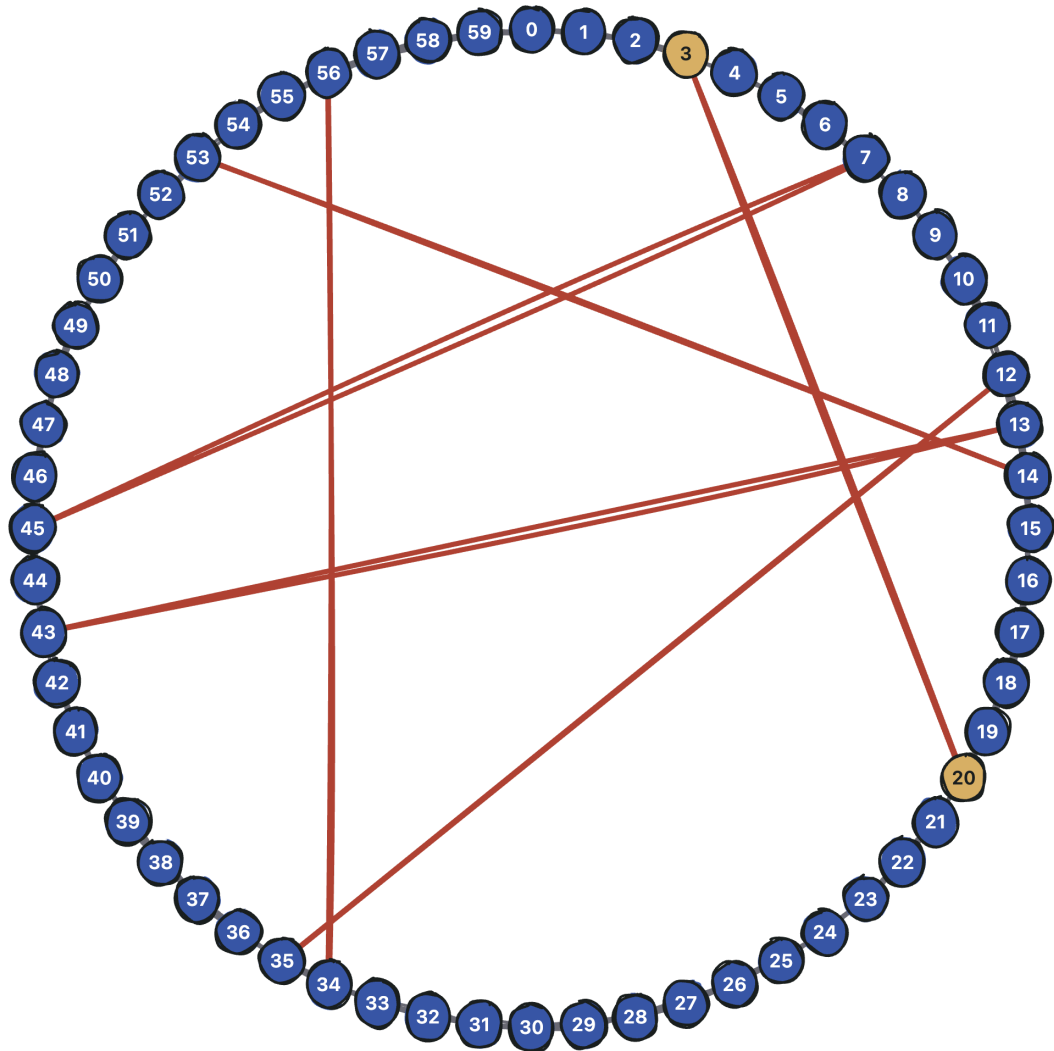
The two numbers are measured on a 200-person ring, averaged over 3 tries. The picture below is a 60-dot sketch of the same rewiring, small enough to see. **Grey lines are the ring's original neighbour links.** **Rust lines are the ones the slider has picked up and reconnected somewhere random.** Move the slider one notch at a time and watch the two numbers above. Take one number at a time: L/L_0 first, then C/C_0 .



links rewired so far 0

6 of the ring's 120 links moved average distance $L = 4.50$ clustering $C = 0.45$

Untouched, this ring has $L = 7.88$ and $C = 0.50$. Grey lines are the ring's original neighbour links. A rust line has been picked up and reconnected to somebody chosen at random. The two amber dots are the ends of the newest one. Step the slider up one at a time and watch the two numbers above. Drag any dot to see what its rust line reaches across.



The Watts–Strogatz recipe (1998)

The dials: p = the fraction of links picked up and reconnected to a random person; L = average distance (typical steps between two people); C = clustering (how often two of your friends know each other). L_0 and C_0 are the same two on the untouched ring, so the ratios read "compared to before". Rewiring about 1% of a ring's links already pulls distance down hard while clustering barely moves. On the 200-person ring in this slider:

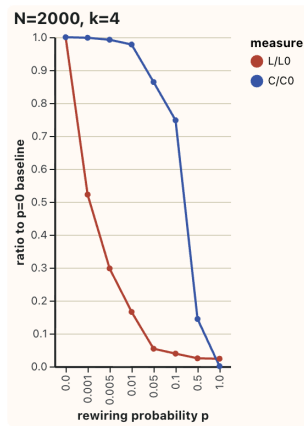
$$p \approx 0.01 \quad \implies \quad \frac{L}{L_0} \approx 0.6 \quad \frac{C}{C_0} \approx 1$$

Those few rewires are the long cables from my drawing: rare enough to keep the communities, long enough to shortcut the ring. 200 people is a small ring, and the effect only sharpens with the crowd: the next cell runs the same sweep at $N = 2000$, where the same 1% takes distance far lower still. That's the small-world recipe, and it's why six degrees works.

Run the same sweep on a much bigger network — $N = 2000$ instead of 200. Fill in the two blanks (each function is named in the comment right above it), then press ► Run. Two lines, both measured against the untouched ring: **rust is L/L_0 , the average distance**; **navy is C/C_0 , the clustering**. Left to right is more rewiring.

```
1 # N is set for you — much bigger than the N=200 you played with.
2 N = 2000
3 k = 4
4 p_values = [0.0, 0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1.0]
5
6 L0 = C0 = None
7 rows = []
8 for p in p_values:
9     G = nx.connected_watts_strogatz_graph(N, k, p, seed=1) if p > 0 else nx.watts_strogatz_graph(N, k, 0)
10    # 1. Measure the average distance of G — nx.average_shortest_path_length
11    L = nx.average_shortest_path_length(G)
12    # 2. Measure the average clustering of G — nx.average_clustering
13    C = nx.average_clustering(G)
14    if p == 0.0:
15        L0, C0 = L, C
16        rows.append({"p": str(p), "L/L0": L / L0, "C/C0": C / C0})
17
18 df = pd.DataFrame(rows).melt("p", var_name="measure", value_name="ratio")
19 alt.Chart(df).mark_line(point=True).encode(
20     x=alt.X("p:0", title="rewiring probability p"),
21     y=alt.Y("ratio:0", title="ratio to p=0 baseline"),
22     color=alt.Color(
23         "measure:N",
24         # Rust = L/L0, navy = C/C0. The domain MUST be bound: without
25         # it Vega sorts the categories and swaps the two colours.
26         scale=alt.Scale(domain=["L/L0", "C/C0"], range=["#B14434", "#3959A6"]),
27     ),
28 ).properties(title=f"N={N}, k={k}")
```

► Run my code



Run it as many times as you like. When it does what you want, press — that is what hands it in and tells your tutor to look.

Send my code to

✓ **Handed in.** Your tutor is reading your code now.

The phase transition, at scale

Same experiment as the slider, but coded and run on a much bigger network — N , the number of people, is 2000 here instead of 200. I swept p (the fraction of links rewired at random) and plotted L/L_0 (average distance, compared to the untouched ring) against C/C_0 (clustering, compared to the untouched ring).

Chapter 5 of 5 — Mastery Check

No new machinery in this final chapter. Just two tests of whether the story holds. First, an AI hands me its analysis of a network. It asks me to sign off on it. Then a friend asks the question this module opened with, and I answer it in my own words.

The analysis I was asked to sign off on

I analyzed your network. Its average clustering coefficient is **0.61**, far higher than a random network's (**0.01**). Therefore this is a small-world network.

Read it as the reviewer who has to sign underneath.

Reviewing an AI's claim

"High clustering $\Rightarrow$ small world" is NOT enough: small-world means BOTH high clustering AND short paths. My counterexample: the ring with nothing rewired ($p = 0$, where p is the fraction of links picked up and reconnected at random) — very clustered, yet enormous distances. So both halves have to be measured: C (clustering — how often two of your friends know each other) and L (average distance — the typical number of steps between two people). Why not just divide one by the other? C is never above 1 and L is never below 1, so C/L is biggest for the network where everyone already knows everyone ($C = 1$, $L = 1$) — the least interesting network there is. Raw numbers certify nothing. The **small-world index** compares both against a random network of the same size instead:

$$\sigma = \frac{C/C_{\text{random}}}{L/L_{\text{random}}} \quad \text{with} \quad C_{\text{random}} \approx \frac{k}{N} \quad L_{\text{random}} \approx \frac{\log N}{\log k}$$

— the two baselines from chapter 3, where N is how many people there are and k how many friends each one has. $\sigma > 1$ means small world: more clustered than random without paying for it in distance. $\sigma \approx 1$ means the network is indistinguishable from a random one.

The whole story, in my own words

Local links make clusters; a few long-range shortcuts exist; those shortcuts pull everyone close. That's how a letter crosses a country in about six steps. And not only a country of people. The same shape shows up wherever wiring is mostly local and expensive to stretch: a brain connects mostly to nearby neurons with a few long fibres crossing it, and the internet is mostly regional with a handful of cables spanning oceans. Both are small worlds for the same reason mine is.

Session record

*Your answer to each question, and every word you typed while working on it — this is what gets reviewed, not the code. Hints are never held against you. **cp0_welcome** · pass (no question — session greeting only)*

My initial guess is that the letter might need 6-8 people to reach the target.

You typed: "My initial guess is that the letter might need 6-8 people to reach the target."

*Tutor's note: Session opened; student volunteered an early guess about the letter's hops before any question was asked. **cp1_milgram** · pass *For the packets that reached the Boston stockbroker, how many people handled one along the way, on average?**

It was very close to my guess. I guessed around 6-8 people, and the actual successful chains used about 5.2 intermediaries on average, so roughly six degrees of separation.

*You chose: "about 6" You typed: "It was very close to my guess. I guessed around 6-8 people, and the actual successful chains used about 5.2 intermediaries on average, so roughly six degrees of separation." Tutor's note: Guessed 6-8; picker chose "about 6"; reaction confirmed the real 5.2-average result matched his guess closely. **cp1_routing** · pass *If I told you, with total certainty, that SOME six-click path connects those two pages somewhere on Wikipedia — would that make the race easy to win? Why or why not?**

Not necessarily. Knowing that a six-click path exists tells me the two pages are close in the network, but it does not tell me which links form that path. At each page I still have many possible links to choose from, and most choices may lead away from the target. So the shortest path can be small even when finding that path using only local information is difficult.

*You typed: "Not necessarily. Knowing that a six-click path exists tells me the two pages are close in the network, but it does not tell me which links form that path. At each page I still have many possible links to choose from, and most choices may lead away from the target. So the shortest path can be small even when finding that path using only local information is difficult." Tutor's note: Articulated precisely that a short path existing differs from finding it with only local info — used Wikirace framing unprompted and correctly. **cp2_distance** · pass *What's the distance from A to D, and then the average over all 6 pairs?**

The distance from A to D is 2, because the shortest route uses two edges. Even though there are multiple possible routes, the shortest-path distance is the minimum number of edges, which is 2. There are 6 unordered pairs among 4 people. Five pairs are directly connected, so their distance is 1, while A-D has distance 2. Therefore the average shortest-path distance is

$1+1+2+1+1+1/6 = \text{approx } 1.17$. So on average, two people in this network are only about 1.17 steps apart.

You typed: "The distance from A to D is 2, because the shortest route uses two edges. Even though there are multiple possible routes, the shortest-path distance is the minimum number of edges, which is 2." · "There are 6 unordered pairs among 4 people. Five pairs are directly connected, so their distance is 1, while A–D has distance 2. Therefore the average shortest-path distance is $1+1+2+1+1+1/6 = \text{approx } 1.17$ So on average, two people in this network are only about 1.17 steps apart." *Tutor's note:* Computed A–D=2 and average $7/6 \approx 1.17$ on first try, with clear reasoning about shortest vs any route and the pair count.

cp2_diameter · pass *Which pair is the single longest trip in this network, and how many edges?*

The pair is A–D, with distance 2. Every other pair is directly connected with distance 1, so A–D is the longest shortest path in the network. Therefore the network's diameter is 2.

You typed: "The pair is A–D, with distance 2. Every other pair is directly connected with distance 1, so A–D is the longest shortest path in the network. Therefore the network's diameter is 2." *Tutor's note:* Named A–D at 2 and even supplied the term "diameter" himself before it was introduced. **cp2_paperwork** · pass *By hand, find the distance for all 10 pairs on the 5-ring, then the average.*

I drew a 5-node ring with nodes A, B, C, D, and E, connected only to their two neighbors: A–B, B–C, C–D, D–E, and E–A. The pairwise distances I got are: A–B = 1, A–C = 2, A–D = 2, A–E = 1, B–C = 1, B–D = 2, B–E = 2, C–D = 1, C–E = 2, and D–E = 1. The total distance is 15 across 10 pairs, so the average path length is $15/10 = 1.5$

You typed: "I drew a 5-node ring with nodes A, B, C, D, and E, connected only to their two neighbors: A–B, B–C, C–D, D–E, and E–A. The pairwise distances I got are: A–B = 1, A–C = 2, A–D = 2, A–E = 1, B–C = 1, B–D = 2, B–E = 2, C–D = 1, C–E = 2, and D–E = 1. The total distance is 15 across 10 pairs, so the average path length is $15/10 = 1.5$ " *Tutor's note:* Photo was uploaded but could not be read on the tutor's side (whiteboard image view unavailable), so student typed the full table: all 10 pair distances correct, average $15/10 = 1.5$. Hand-worked page was still produced and submitted. **cp3_clustering** · pass *How many friendships actually exist among A's 5 friends, and what fraction is that?*

There are 2 actual friendships among Alice's five friends. Since 10 friendships are possible, Alice's local clustering is $2/10 = 0.2$.

You typed: "Alice has the same number of friends in both cases, so her degree is the same. What changes is the amount of connection among her five friends. In the first case, none of Alice's friends know each other, so there are no links among her neighbors. In the second case, every one of her friends knows every other friend, so her neighborhood is fully

connected. This is capturing local clustering: how tightly connected a node's neighbors are to each other." · "I would count how many connections exist between Alice's friends and compare that with how many connections could possibly exist between them. If none of her friends know each other, the value is 0. If all of them know each other, the value is 1." ·

"There are 2 actual friendships among Alice's five friends. Since 10 friendships are possible, Alice's local clustering is $2/10=0.2$." *Tutor's note:* Identified the difference as lines between friends and named clustering; counted 2 of 10 actual friendships and computed 0.2 correctly in one breath. **cp3_average** · pass *B's score, then C/E/F's scores, then D's (with one friend), then average all six.*

B knows A and F. Those two are also connected, so B's score is 1.0. C, E, and F all have the same score as B: 1.0, because in each case their two friends are also connected. D has only one friend, so there is no pair of D's friends that could be connected — zero possible links to evaluate. The average local clustering coefficient is 0.7.

You typed: "B knows A and F. Those two are also connected to each other, so B's two neighbors form the only possible connection between them. So B's local clustering score is 1.0 or 100%" · "C, E, and F all have the same score as B: 1.0, because in each case their two friends are also connected to each other." · "D has only one friend, so there is no pair of D's friends that could be connected. There are zero possible friend-to-friend links to evaluate." · "The average local clustering coefficient is 0.7. I added the six individual scores and divided by six." *Tutor's note:* Scored B, C, E, F = 1 each, recognized D has zero pairs to check (hence 0 by convention), and averaged to 0.70 correctly. **cp3_global_clustering** · pass *Why does the global measure land near Alice's 0.2 while the average said 0.70?*

The global measure gives more weight to people who center many triplets. Alice alone accounts for 10 of the 14 triplets, and her neighborhood is only weakly clustered, so her low value has a much bigger influence. In the average local clustering, every person counts equally, which is why that value was much higher at 0.70.

You typed: "A closed triplet" · "There are 2 triangles in the whole network. A–B–F and A–C–E. Each of those trios has all three connections present." · "B, C, E, and F each center 1 triplet, D centers 0, and Alice centers 10, so the network has 14 triplets altogether." · "The network's global clustering coefficient is about 0.43. That is lower than the average local clustering of 0.70, so the two ways of measuring clustering do not have to agree." · "The global measure gives more weight to people who center many triplets. Alice alone accounts for 10 of the 14 triplets, and her neighborhood is only weakly clustered, so her low value has a much bigger influence. In the average local clustering, every person counts equally, which is why that value was much higher at 0.70." *Tutor's note:* Named closed triplet, found 2 triangles, totaled 14 triplets ($10+1+1+1+1+0$), computed 0.43, and — unprompted and precisely — explained the hub-weighting reason for the 0.70 vs 0.43 gap.

cp4_shortcut_drawing · pass *On 8 dots the gap is small. What would you expect on a ring of 800?*

On a ring of 800 nodes, I would expect the long cable to make a much bigger difference. Without shortcuts, many pairs of nodes are very far apart because you have to travel around the ring step by step. A long-range cable connects distant parts of the network directly, so it can shorten the routes for many different pairs at once. As the ring gets larger, that kind of shortcut should become much more valuable than a short cable between already-nearby nodes.

You typed: "I connected A and E, the two dots directly opposite each other on the 8-node ring. I chose them because they were originally 4 steps apart, so I expected that shortcut to reduce many shortest-path distances across the network." · "On a ring of 800 nodes, I would expect the long cable to make a much bigger difference. Without shortcuts, many pairs of nodes are very far apart because you have to travel around the ring step by step. A long-range cable connects distant parts of the network directly, so it can shorten the routes for many different pairs at once. As the ring gets larger, that kind of shortcut should become much more valuable than a short cable between already-nearby nodes." *Tutor's note:* Connected A–E straight across (the long-range choice) with sound reasoning; then predicted the gap widens on a bigger ring, correctly seeing long cables save many steps at once. **cp5_ring_formula** · pass *Does clustering depend on N, and how does L relate to N and k?*

No, clustering depends on k not N. L grows with N and shrinks with k, roughly $N/(2k)$.

You typed: "No, it shows 3 connected pairs, not 2. I missed the connection between 11 and 1 across node 0's neighborhood. So for K=4, node 0 has 4 friends and 3 actual links among those friends." · "Node 0's clustering coefficient is $3/6 = 0.5$, because 3 of the 6 possible friendships among its four friends actually exist." · "No. For this regular k=4 ring, node 0 still has the same four friends, and the same 3 of 6 possible connections among those friends are present. So its clustering stays $3/6=0.5$. In this setup, the local clustering depends on k, not on the total number of people N." · "It would take about 250 hops to reach the person directly opposite me. That shows that, unlike clustering, path length grows with the size N of the ring." · "The average path length L grows as N grows, because a larger ring means people are spread farther apart. It shrinks as K grows, because having more friends lets each hop cover more of the ring. So, roughly, L increases with N and decreases with K." *Tutor's note:* Worked k=2 (no triangles) and k=4 (caught the missed 11-1 pair via the count box, $3/6=0.5$), stated C depends only on k not N, got 250 hops for $N=1000/k=4$, and wrote $L \approx N/(2k)$. Even generalized $C(k)=3(k-2)/(4(k-1))$ on his own page. Photo read successfully this time. **cp5_tension** · pass *Ring world and random world — what do your C and L formulas say about each, and which world do you live in?*

I think the real world is a mix of the two. My friends often know each other, so clustering is high, but I also have some connections to people far outside my local group, which create shortcuts. So real social networks can have both high clustering and short average path lengths.

You typed: "Yes. If everyone still knows only their nearest few neighbors on the ring, the people around me will still tend to know each other. The clustering coefficient stays determined by k , not by the total population size. So making the country much larger does not by itself reduce local clustering." · "It would take a very long time compared with a small network. Since L grows with N , making the ring country-sized makes typical routes huge unless K also grows. So the network can have high clustering but still have very long paths." · "L becomes small, because random long-range connections create shortcuts across the country, so you can reach distant people in relatively few steps. C becomes low, because your friends are scattered randomly and are therefore unlikely to also know each other." · "I think the real world is a mix of the two. My friends often know each other, so clustering is high, but I also have some connections to people far outside my local group, which create shortcuts. So real social networks can have both high clustering and short average path lengths." *Tutor's note:* Correctly read both worlds off his own formulas — ring: high C , L grows; random: L small, C low — and concluded the real world is a mix with both, precisely the instinct the next chapter builds on. **cp6_watts_strogatz** · pass *Which single notch drops L/L_0 the most, and what is C/C_0 at that notch?*

The biggest single-notch drop in normalized distance happens at $p = 0.05$, where L/L_0 falls from 0.61 to 0.29, a drop of 0.32. At the same point, $C/C_0 = 0.84$, which is still close to 1. So only a small amount of rewiring greatly shortens paths while preserving most of the original clustering.

You chose: "distance drops a lot, clustering barely changes" *You typed:* "The biggest single-notch drop in normalized distance happens at $p = 0.05$, where L/L_0 falls from 0.61 to 0.29, a drop of 0.32. At the same point, $C/C_0 = 0.84$, which is still close to 1. So only a small amount of rewiring greatly shortens paths while preserving most of the original clustering." *Tutor's note:* Prediction correct upfront; then read the steepest drop at $p = 0.05$ ($0.61 \rightarrow 0.29$, drop 0.32) with $C/C_0 = 0.84$ still high, and reconciled it to the formulas unprompted. **cp6_large_n_experiment** · pass *At which p does L/L_0 drop steepest, what is C/C_0 there, and is that p smaller than on the 200-person ring?*

The steepest drop in L/L_0 happens at about $p = 0.001$, where the normalized distance falls from 1.00 to roughly 0.52. At that same point, C/C_0 is still almost 1. Compared with the 200-person ring, this happens at a smaller rewiring probability, so the small-world effect appears even earlier in the larger network.

You typed: "The steepest drop in L/L_0 happens at about $p = 0.001$, where the normalized distance falls from 1.00 to roughly 0.52. At that same point, C/C_0 is still almost 1. Compared

with the 200-person ring, this happens at a smaller rewiring probability, so the small-world effect appears even earlier in the larger network." *Tutor's note:* Filled both blanks correctly, ran clean, and read the steepest drop at $p=0.001$ with $C/C_0 \approx 1$, correctly naming it smaller than the $N=200$ notch — the graded half. **cp7_redteam** · pass *An AI claims small-world from clustering 0.61 vs 0.01 alone. You're the reviewer who has to sign off. Do you?*

I would not approve that conclusion yet. High clustering compared with a random network is only one half of the small-world property. We also need to check whether the network's average path length is close to the random network's path length. A small-world network should have both high clustering and short paths, so the assistant's evidence is incomplete.

You typed: "I would not approve that conclusion yet. High clustering compared with a random network is only one half of the small-world property. We also need to check whether the network's average path length is close to the random network's path length. A small-world network should have both high clustering and short paths, so the assistant's evidence is incomplete." *Tutor's note:* Refused to sign off, correctly naming that path length was never measured and that small-world needs both high clustering AND short paths — even gestured at the random-baseline comparison (the bonus). **cp8_wrapup** · pass *How can a letter cross a whole country in about six steps?*

A country can still feel "small" because most people belong to close local groups, but a few connections reach far outside those groups. Those long-distance connections act like shortcuts, so a message can jump across large parts of the network very quickly. That is why two people who seem very far apart can still be connected through only a few other people.

You typed: "A country can still feel "small" because most people belong to close local groups, but a few connections reach far outside those groups. Those long-distance connections act like shortcuts, so a message can jump across large parts of the network very quickly. That is why two people who seem very far apart can still be connected through only a few other people." *Tutor's note:* Explained the phenomenon end to end in plain words, hitting all three beats — local clusters, a few long-range shortcuts, shortcuts pulling everyone close — with no jargon and no hints.

 **Stuck with your tutor?** If the two of you are going in circles — you think an answer should count, you want a fresh try, or you'd rather move on — say so here and press the button. A second, stronger model reads the whole situation and makes a call your tutor has to follow. Using it is never held against you.

What's going on? e.g. "I answered this and I think it should count", "I'd like a fresh try at this one", "we're going in circles — let's move on"

 **Tutor gets stuck — call**